

ARSHIYA KHAN

SUMMARY

Ph.D. candidate specializing in AI-driven cybersecurity and large-scale threat detection systems. Over 6 years of experience in designing and deploying machine learning pipelines to detect anomalies, vulnerabilities, and adversarial behaviors.

EDUCATION

Ph.D. Electrical and Computer Engineering

2019 - 2026 (*expected*)

University of Delaware

Newark, DE

- Advisors: Dr. Xing Gao, Dr. Chase Cotton
- Dissertation: Red-teaming LLMs: Bypassing security guardrails of AI-driven systems by hiding malicious intent in prompts. Converted unsafe prompt to safe by token manipulation.

M.S. Cybersecurity

2017 - 2019

University of Delaware

Newark, DE

- Advisor: Dr. Chase Cotton
- Thesis: Taxonomy of network traffic to perform ML-based anomaly detection for new attack signals

TECHNICAL SKILLS

Machine Learning: LLM safety and alignment, Malware detection, Vulnerability repair, Statistical analyses, Natural language processing (NLP), Reinforcement learning

Network Security: Packet analysis, Network traffic analysis, Pen testing, Reverse engineering, System hardening, Fuzzing, Cryptography

ML Frameworks & Tools: Torch, TensorFlow, Scikit-learn, Keras, Hugging Face, NumPy, Pandas, Scipy, OpenCV, PIL

Programming Languages: Python, R, MATLAB, Bash, Java, C++, MySQL

PUBLICATIONS

1. **Code Vulnerability Repair with LLM using Context-Aware Prompt Tuning**
IEEE Security and Privacy Workshops (SPW) 2025
Improved buffer overflow vulnerability repair by 74% on GitHub Copilot. Used prompt tuning to introduce security and code knowledge to code-generating LLMs.
2. **Ensemble Defense System: A Hybrid IDS Approach for Effective Cyber Threat Detection**
IEEE International Telecommunication Networks and Applications Conference 2023
Developed a hybrid intrusion detection model by integrating signature-based and anomaly-based IDS tools. Designed an interactive tracker for tracking and visualizing threat vectors.
3. **Efficient attack detection in IoT devices using feature engineering-less machine learning**
International Journal of Computer Science and Information Technology 2022
Designed a lightweight malware detection model that removed the feature-engineering process from the ML pipeline. Deployable on battery-powered, low-memory IoT devices.
4. **Detecting Attacks on IoT Devices using Featureless 1D-CNN**
IEEE International Conference on Cyber Security and Resilience (CSR) 2021
Used pcap files as raw time-series signals to train a 1D-CNN for malware detection. Emphasize the role of headers in malware detection.
5. **A Feature Taxonomy for Network Traffic**
University of Delaware, Department of Electrical and Computer Engineering 2019
Developed a taxonomy model to detect anomalous activities detected in packets, flows, and sessions. Performed extensive analysis of ML-based anomaly detection techniques based on raw, behavioral, and temporal features. Highlighted anomalies detected by network traffic scanners such as Zeek and Wireshark.

PROJECTS

1. **JPMC Data for Good Hackathon 2021**
Led a team of 4 to design a forecasting model with Linear Regression with a focus on sparse data handling.
2. **Twitter Bot Detection and Classification 2019**
Built a Multinomial Naive Bayes model, achieving 91% accuracy on detecting harmful bot accounts on Twitter and classifying them into political, social, and personal harm categories.

WORK EXPERIENCE

Graduate Teaching Assistant | University of Delaware 2019 – Present

- Developed and taught a graduate-level class, “Topics in Cybersecurity”, on implementing supervised ML techniques for anomaly detection in large-scale data sources like packet captures or compiled network security logs from Zeek.
- Collaborated with professors to guide undergraduate, graduate, and mid-career professionals. Teaching Assistant for CPEG694 System Hardening and Protection, TOPICS19F Topics in Cybersecurity, CPEG665 Introduction to Cybersecurity, CPEG697 Advanced Cybersecurity.
- Received recognition for creating a supportive and inclusive environment in the classroom as a TA.

Backend & Analytics Developer | Women Plus Blockchain 2018.06 - 2018.08

- Built 50 web crawlers to collect data from blockchain news outlets; implemented a scalable pipeline on AWS console to filter relevant articles using SEO-based screening
- Boosted subscriber growth by 17% by automating user engagement metrics on Google Analytics dashboard in 4 months

Systems Engineer | Tata Consultancy Services Ltd 2014.09 - 2017.04

- Led an application team of 5 members to support and maintain a telecom banking application, serving daily transactions
- Deployed changes to the production environment after application testing, resolved critical incidents, maintaining 99% uptime SLA
- Cut reporting costs by 90% by automating daily business intelligence report generation using VBA

TEAMWORK EXPERIENCE

- Led peer mentoring organization EmPOWER at University of Delaware 2021-2023
- Associate, Hispanic Latino Graduate Student Organization 2022
- Treasurer, Indian Graduate Student Organization 2021
- UMBC CyberDawgs CTF at the University of Maryland 2019